

# McNemar's Test

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## Introduction

When the same units are measured on a binary outcome twice (before/after, two tests, two raters), the observations are paired and a standard two-proportion test is invalid. McNemar's test analyses the 2x2 table of paired responses, using only the **discordant pairs** – those where the two measurements differ.

## Prerequisites

Contingency tables, paired design.

## Theory

Arrange paired binary data in a 2x2 table:

|           | Test 2: +        | Test 2: -        |
|-----------|------------------|------------------|
| Test 1: + | $a$ (agree)      | $b$ (discordant) |
| Test 1: - | $c$ (discordant) | $d$ (agree)      |

McNemar's null: the probability of switching from + to - equals the probability of switching from - to +; equivalently, the marginal prevalences are equal.

Test statistic:

$$\chi^2_{\text{McNemar}} = \frac{(b - c)^2}{b + c} \sim \chi^2_1 \quad \text{under } H_0.$$

For small  $b + c$ , an exact binomial test on  $b$  out of  $b + c$  with  $p_0 = 0.5$  is preferred.

The concordant cells  $a$  and  $d$  do not enter the test; only the discordant  $b$  and  $c$  do.

## Assumptions

- Paired binary data.
- Independent pairs.
- $b + c$  sufficiently large ( $\geq 25$ ) for the chi-squared approximation; otherwise exact.

## R Implementation

```
# Paired data: a diagnostic test evaluated before and after training
tab <- matrix(c(45, 12,
               25, 18), nrow = 2, byrow = TRUE,
              dimnames = list(Before = c("+", "-"),
                              After  = c("+", "-")))
tab
```

```

# McNemar (with continuity correction disabled)
mcnemar.test(tab, correct = FALSE)

# Exact McNemar via exact2x2
library(exact2x2)
mcnemar.exact(tab)

# Manual McNemar
b <- tab[1, 2]; c <- tab[2, 1]
chi <- (b - c)^2 / (b + c)
p_val <- 1 - pchisq(chi, df = 1)
c(chi = chi, p = p_val)

```

## Output & Results

McNemar's Chi-squared test  
 McNemar's chi-squared = 4.568, df = 1, p-value = 0.0326

Exact McNemar test (exact2x2):  
 proportion +1 to -1: 0.324  
 95% CI: 0.037 to 0.586  
 p-value = 0.041

Discordant pairs: 12 (before+ / after-) vs. 25 (before- / after+). McNemar  $\chi^2(1) = 4.57$ ,  $p = 0.033$ ; exact  $p = 0.041$ .

## Interpretation

Reporting: “Of 37 discordant pairs, 25 switched from negative to positive and 12 switched in the opposite direction; McNemar’s exact test gave  $p = 0.041$ , indicating a significant change after training.”

## Practical Tips

- Report only the discordant counts and the p-value; concordant counts are irrelevant to McNemar.
- For  $b + c < 25$ , use the exact version.
- McNemar’s odds ratio is  $b/c$ ; its CI is computed by inverting the exact test.
- For more than two categories per assessment (paired, more than two outcomes), use Bowker’s symmetry test or Stuart-Maxwell.
- Stratified McNemar: use conditional logistic regression (`survival::clogit`) for a set of matched pairs with covariates.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

## See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests

- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests